Last time

- Recap
- Example: Airplane
- Simulation

Today

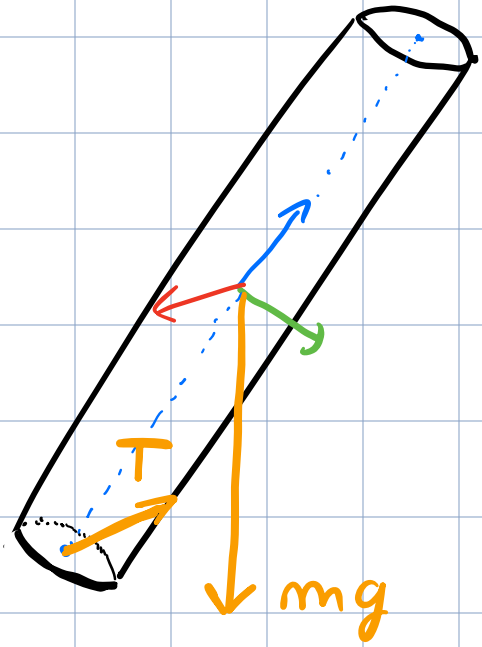
- Example 1: Rocket
- Example 2 and 3: Unconventional vehicle

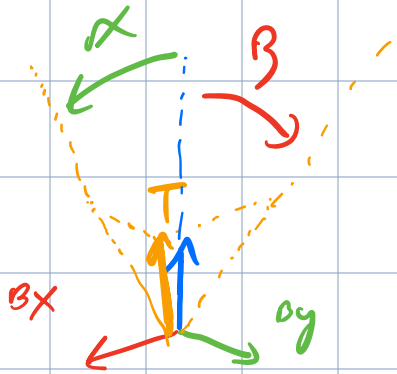
Lots of systems can be modeled with Newton-Euler:

$$x = \begin{bmatrix} {}^{N}r & {}^{N}q^{B} & {}^{B}v & {}^{B}\omega \end{bmatrix}^{T}$$

$$\dot{x} = \begin{bmatrix} Q\,{}^{B}v \\ \frac{1}{2}G(q)\,{}^{B}\omega \\ \frac{1}{m}\,{}^{B}F - {}^{B}\omega \times {}^{B}v \\ J^{-1}\left({}^{B}\tau - {}^{B}\omega \times J\,{}^{B}\omega\right) \end{bmatrix}$$

EXAMPLE: THRUST-VECTOR ROCKET





Longitudinal forces

$$\Rightarrow {}^{B}F = {}^{B}F_T + Q^T \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix}$$

thrust — weight

$$T_0 = \begin{bmatrix} 0 \\ 0 \\ T \end{bmatrix} \qquad R_y(\alpha) = \begin{bmatrix} \cos\alpha & 0 & \sin\alpha \\ 0 & 1 & 0 \\ -\sin\alpha & 0 & \cos\alpha \end{bmatrix}$$

$$T_1 = R_y(\alpha)\, T_0 = T \begin{bmatrix} \sin\alpha \\ 0 \\ \cos\alpha \end{bmatrix}$$

$$T_2 = R_x(-\beta)\, T_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\beta & \sin\beta \\ 0 & -\sin\beta & \cos\beta \end{bmatrix} T_1$$

$${}^{B}F_T = T_2 = T \begin{bmatrix} \sin\alpha \\ \sin\beta\cos\alpha \\ \cos\beta\cos\alpha \end{bmatrix}$$

$$\boxed{{}^{B}F = T \begin{bmatrix} \sin\alpha \\ \sin\beta\cos\alpha \\ \cos\beta\cos\alpha \end{bmatrix} + Q^T \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix}}$$

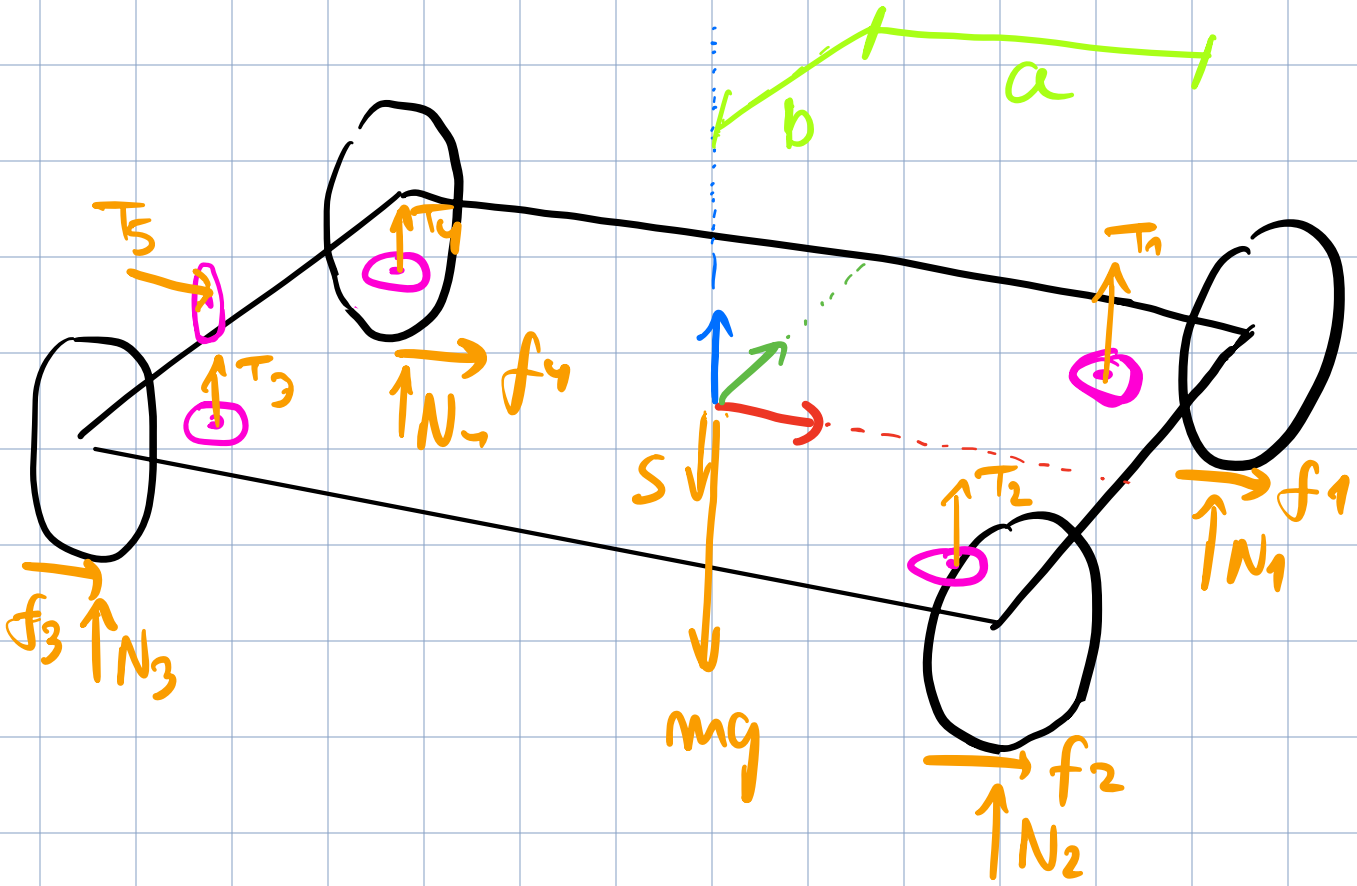
Angular forces (torques)

$${}^{B}Z = {}^{B}r_T \times {}^{B}F_T = \begin{bmatrix} 0 \\ 0 \\ -\ell \end{bmatrix} \times {}^{B}F_T$$

$$\boxed{{}^{B}Z = \begin{bmatrix} \ell T \sin\beta\cos\alpha \\ -\ell T \sin\alpha \\ 0 \end{bmatrix}}$$

We can't spin over longitudinal axis

EXAMPLE 2: FLYING-CAR



Longitudinal forces

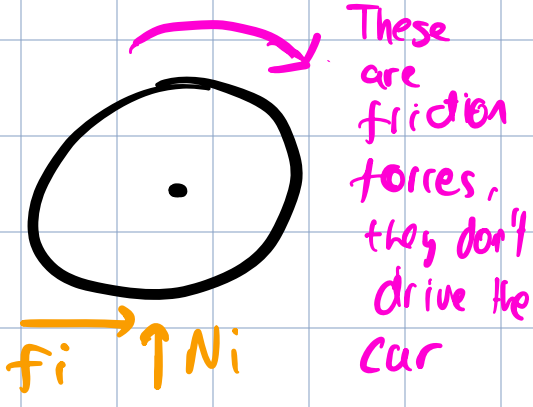
$${}^3F = {}^3F_{TF} + {}^3F_W + {}^3F_{GT} + {}^3F_S$$

TF → Tire forces
W → Weight
GT → Gas thrusters
S → Suction

$$(TF) \rightarrow {}^3F_{TF} = Q^T(N_i \lambda_i + f_i t_i)$$

$$(i = 1,2,3,4)$$

$$f_i \leq \mu N_i$$



$$(W) \rightarrow {}^{B}F_W = Q^T \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix}$$

$$(GT) \rightarrow {}^{B}F_{GT} = \begin{bmatrix} T_5 \\ 0 \\ T_1 + T_2 + T_3 + T_4 \end{bmatrix}$$

These fly | drive | the car!!

$$(S) \rightarrow {}^{3}F_S = \begin{bmatrix} 0 \\ 0 \\ -S \end{bmatrix}$$

$$\Rightarrow \boxed{{}^{B}F = \begin{bmatrix} T_5 \\ 0 \\ \sum_{i=1}^{4} T_i - S \end{bmatrix} + \sum_{i=1}^{4} Q^T \left(N_i n_i + f_i t_i \right)}$$

Angular forces (torques)

$${}^{B}\tau = {}^{B}r \times {}^{B}F$$

$$(TF) \rightarrow {}^{B}\tau_{TF} = \sum_{i=1}^{4} {}^{B}r_i \times (N_i n_i + f_i t_i)$$

$$(W) \rightarrow {}^{B}\tau_W = 0$$

$$(GT) \rightarrow {}^{B}\tau_{GT} = \begin{bmatrix} b(T_1 + T_4 - T_2 - T_3) \\ a(T_3 + T_4 - T_1 - T_2) \\ 0 \end{bmatrix}$$

$$(S) \rightarrow {}^{B}\tau_S = 0$$

$$\Rightarrow {}^{B}\tau = \begin{bmatrix} b(T_1 + T_4 - T_2 - T_3) \\ a(T_3 + T_4 - T_1 - T_2) \\ 0 \end{bmatrix} + \sum_{i=1}^{4} {}^{B}r_i \times (N_i n_i + f_i t_i)$$